

DSA0405-Fundamental of Data Science

CO3-AT1 - Case Study

UNIT -3

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Case Study 6 — Detecting an Unusual Customer:

A shopping website finds that the average amount spent by customers is ₹2,000, with a standard deviation of ₹400. A particular customer spends ₹3,200.

SOLUTION:

Given Data:

Mean spending, $\mu = ₹2,000$

Standard deviation, $\sigma = ₹400$

Spending of the particular customer, $x = ₹3,200$

Question 1: Calculate the customer's z-score.

$$z = (x - \mu) / \sigma$$

Where:

z = the z-score (number of standard deviations a value is from the mean)

x = the observed data value = 3200

μ = the population mean = 2000

σ = the population standard deviation = 400

Step 1: Find the numerator ($x - \mu$).

$$x - \mu = 3200 - 2000 = 1200$$

Step 2: Divide the numerator by σ .

$$z = 1200 / 400$$

Step 3: Simplify the division.

$$z = 3$$

Final Answer: $z = 3$

Question 2: Is the customer's spending unusually high compared with the average?

Rule: If $|z| > 2$ (or 3), the value is considered statistically unusual.

Step 1: Take the absolute value of the z-score calculated above.

$$|z| = |3| = 3$$

Step 2: Compare $|z|$ with the threshold value 2 (commonly used cut-off for 'unusual').

$3 > 2 \rightarrow$ condition satisfied

Final Answer: Yes, the spending of ₹3,200 is unusually high, since $z = 3$ places it 3 standard deviations above the mean.

Question 3: What would a negative z-score mean in this context?

$$z = (x - \mu) / \sigma$$

Step 1: Set the z formula less than 0 and solve the inequality for x.

$$(x - \mu) / \sigma < 0$$

Step 2: Since σ (400) is always positive, the sign of z depends only on $(x - \mu)$.

$$(x - \mu) < 0 \Rightarrow x < \mu$$

Step 3: Substitute $\mu = 2000$.

$$x < 2000$$

Final Answer: A negative z-score would mean the customer spent less than the average of ₹2,000.

Question 4: If another customer has a z-score of -1.5, what does it tell you?

$$x = \mu + z \cdot \sigma$$

Where:

x = spending amount to be found

μ = mean = 2000

z = given z-score = -1.5

σ = standard deviation = 400

Step 1: Multiply z by σ .

$$z \times \sigma = -1.5 \times 400 = -600$$

Step 2: Add this result to μ .

$$x = 2000 + (-600)$$

Step 3: Simplify.

$$x = 1400$$

Final Answer: The customer spent ₹1,400, which is 1.5 standard deviations below the mean — lower than typical spending, but not extreme enough to be called an outlier ($|z| < 2$).

Case Study 7 — Delivery Time Analysis

A delivery company records delivery times (in minutes): 25, 27, 28, 29, 30, 30, 31, 32, 34, 60

SOLUTION:

Data (delivery times in minutes), $n = 10$:

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Question 1: Calculate the mean and median.

$$\text{Mean} = \Sigma xi / n$$

Where:

Σxi = sum of all data values

n = number of data values = 10

Step 1: Add the values in pairs to find Σxi .

$$25 + 27 = 52$$

$$52 + 28 = 80$$

$$80 + 29 = 109$$

$$109 + 30 = 139$$

$$139 + 30 = 169$$

$$169 + 31 = 200$$

$$200 + 32 = 232$$

$$232 + 34 = 266$$

$$266 + 60 = 326$$

Step 2: Divide the total by n .

$$\text{Mean} = 326 / 10 = 32.6$$

Final Answer: Mean = 32.6 minutes

$$\text{Median } (n \text{ even}) = [(n/2)\text{-th value} + (n/2 + 1)\text{-th value}] / 2$$

Step 1: $n = 10$, so $n/2 = 5$ th value and $n/2+1 = 6$ th value.

5th value = 30, 6th value = 30

Step 2: Average the two middle values.

Median = $(30 + 30) / 2 = 60 / 2 = 30$

Final Answer: Median = 30 minutes

Question 2: Find Q1 and Q3.

$Q1$ = median of the lower half of data; $Q3$ = median of the upper half of data

Step 1: Split the ordered data into two halves of 5 values each (excluding overall median position).

Lower half: 25, 27, 28, 29, 30

Upper half: 30, 31, 32, 34, 60

Step 2: Find the median of the lower half (middle of 5 values) → $Q1$.

Lower half ordered: 25, 27, 28, 29, 30 → middle value = 28

Final Answer: $Q1 = 28$ minutes

Step 3: Find the median of the upper half (middle of 5 values) → $Q3$.

Upper half ordered: 30, 31, 32, 34, 60 → middle value = 32

Final Answer: $Q3 = 32$ minutes

Question 3: Calculate the IQR.

$IQR = Q3 - Q1$

Step 1: Substitute $Q3 = 32$ and $Q1 = 28$.

$IQR = 32 - 28$

Step 2: Simplify.

$IQR = 4$

Final Answer: $IQR = 4$ minutes

Question 4: Check for outliers using the $1.5 \times \text{IQR}$ rule.

$$\text{Lower Fence} = Q1 - 1.5 \times \text{IQR} \quad \text{Upper Fence} = Q3 + 1.5 \times \text{IQR}$$

Step 1: Compute $1.5 \times \text{IQR}$.

$$1.5 \times \text{IQR} = 1.5 \times 4 = 6$$

Step 2: Compute the lower fence.

$$\text{Lower Fence} = 28 - 6 = 22$$

Step 3: Compute the upper fence.

$$\text{Upper Fence} = 32 + 6 = 38$$

Step 4: Compare every data value against the fences [22, 38].

All values 25–34 lie inside [22, 38]. The value 60 lies outside ($60 > 38$).

Final Answer: 60 is an outlier; no other values are outliers.

Question 5: Explain whether the mean or median gives a better description of the typical delivery time.

Step 1: Compare mean and median computed above.

$$\text{Mean} = 32.6, \text{Median} = 30$$

Step 2: Note the effect of the outlier (60) on each measure.

The outlier raises the mean above most of the data ($32.6 > 8$ of the 10 values), while the median (30) stays close to the bulk of the data.

Final Answer: The median (30 minutes) better represents the typical delivery time, since it is not distorted by the outlier.

Case Study 8 — Two Cities' Temperatures

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

SOLUTION:

Given Data

City A (°C), $n = 7$: 28, 29, 30, 30, 31, 30, 32

City B (°C), $n = 7$: 20, 25, 30, 35, 30, 25, 45

Question 1: Calculate the mean temperature for each city.

$$\text{Mean} = \Sigma xi / n$$

Step 1 (City A): Add all 7 values.

$$28 + 29 + 30 + 30 + 31 + 30 + 32 = 210$$

Step 2 (City A): Divide by $n = 7$.

$$\text{Mean}(A) = 210 / 7 = 30$$

Final Answer: Mean of City A = 30°C

Step 1 (City B): Add all 7 values.

$$20 + 25 + 30 + 35 + 30 + 25 + 45 = 210$$

Step 2 (City B): Divide by $n = 7$.

$$\text{Mean}(B) = 210 / 7 = 30$$

Final Answer: Mean of City B = 30°C

Question 2: Calculate the variance for each city.

$$\sigma^2 = \Sigma (xi - \mu)^2 / n$$

Where:

xi = each individual data value

μ = the mean of the data set

$(x_i - \mu)$ = deviation of each value from the mean

n = number of data values

Step 1 (City A): Find the deviation and squared deviation of each value from $\mu = 30$.

x_i	$x_i - \mu$	$(x_i - \mu)^2$
28	-2	4
29	-1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

Step 2 (City A): Sum the squared deviations.

$$\Sigma(x_i - \mu)^2 = 4 + 1 + 0 + 0 + 1 + 0 + 4 = 10$$

Step 3 (City A): Divide by $n = 7$.

$$\sigma^2(A) = 10 / 7 = 1.43$$

Final Answer: Variance of City A = 1.43

Step 1 (City B): Find the deviation and squared deviation of each value from $\mu = 30$.

x_i	$x_i - \mu$	$(x_i - \mu)^2$
20	-10	100
25	-5	25
30	0	0
35	5	25
30	0	0
25	-5	25
45	15	225

Step 2 (City B): Sum the squared deviations.

$$\Sigma(xi - \mu)^2 = 100+25+0+25+0+25+225 = 400$$

Step 3 (City B): Divide by $n = 7$.

$$\sigma^2(B) = 400 / 7 = 57.14$$

Final Answer: Variance of City B = 57.14

Question 3: Which city has greater temperature variability?

Step 1: Compare the two variances calculated above.

$$\sigma^2(A) = 1.43 \quad \text{vs} \quad \sigma^2(B) = 57.14$$

Step 2: The larger variance indicates greater spread of data around the mean.

$$57.14 > 1.43$$

Final Answer: City B has far greater temperature variability.

Question 4: Why can two datasets have the same mean but different standard deviations?

$$\text{Mean} = \Sigma xi / n \quad (\text{location measure}) \quad \sigma = \sqrt{[\Sigma(xi - \mu)^2 / n]} \quad (\text{spread measure})$$

Step 1: Note that both formulas use different information from the data.

Mean uses only the sum of values; standard deviation uses the squared distance of each value from the mean.

Step 2: Apply this to City A and City B: both have $\Sigma xi = 210$ giving $\mu = 30$, but the individual values differ greatly in how far they lie from 30.

City A values stay within ± 2 of 30; City B values range from -10 to $+15$ relative to 30.

Final Answer: Two datasets can share the same mean (average level) while having different standard deviations (different spread), because the mean and standard deviation measure different properties of the data.

Question 5: Which city has more predictable temperatures?

Step 1: Predictability corresponds to lower variability, i.e. lower σ^2 .

$$\sigma^2(A) = 1.43 \text{ is much smaller than } \sigma^2(B) = 57.14$$

Final Answer: City A has more predictable temperatures.

Case Study 9 — Data Center CPU Usage

A data scientist records CPU usage percentages: 40, 42, 45, 45, 46, 48, 50, 52, 55, 90

SOLUTION:

Data (CPU usage %), $n = 10$:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Question 1: Calculate mean, median, and mode.

$$\text{Mean} = \sum xi / n$$

Step 1: Add the values in pairs.

$$40 + 42 = 82$$

$$82 + 45 = 127$$

$$127 + 45 = 172$$

$$172 + 46 = 218$$

$$218 + 48 = 266$$

$$266 + 50 = 316$$

$$316 + 52 = 368$$

$$368 + 55 = 423$$

$$423 + 90 = 513$$

Step 2: Divide by $n = 10$.

$$\text{Mean} = 513 / 10 = 51.3$$

Final Answer: Mean = 51.3

$$\text{Median } (n \text{ even}) = [(n/2)\text{-th value} + (n/2 + 1)\text{-th value}] / 2$$

Step 1: $n = 10$, so use 5th and 6th values: 46 and 48.

Step 2: Average them.

$$\text{Median} = (46 + 48) / 2 = 94 / 2 = 47$$

Final Answer: Median = 47

Step 1: Mode = the value that occurs most frequently.

45 occurs twice; all other values occur once.

Final Answer: Mode = 45

Question 2: Calculate the range.

Range = Maximum value – Minimum value

Step 1: Identify maximum and minimum.

Maximum = 90, Minimum = 40

Step 2: Subtract.

Range = $90 - 40 = 50$

Final Answer: Range = 50

Question 3: Find Q1, Q3 and IQR.

$Q1 = \text{median of lower half}$; $Q3 = \text{median of upper half}$; $IQR = Q3 - Q1$

Step 1: Split data into lower and upper halves of 5 values each.

Lower half: 40, 42, 45, 45, 46

Upper half: 48, 50, 52, 55, 90

Step 2: Find the middle value of each half.

$Q1 = \text{middle of lower half} = 45$

$Q3 = \text{middle of upper half} = 52$

Step 3: Compute IQR.

$IQR = Q3 - Q1 = 52 - 45 = 7$

Final Answer: $Q1 = 45$, $Q3 = 52$, $IQR = 7$

Question 4: Identify possible outliers.

$\text{Lower Fence} = Q1 - 1.5 \times IQR$ $\text{Upper Fence} = Q3 + 1.5 \times IQR$

Step 1: Compute $1.5 \times \text{IQR}$.

$$1.5 \times \text{IQR} = 1.5 \times 7 = 10.5$$

Step 2: Compute the lower fence.

$$\text{Lower Fence} = 45 - 10.5 = 34.5$$

Step 3: Compute the upper fence.

$$\text{Upper Fence} = 52 + 10.5 = 62.5$$

Step 4: Compare each data value with the fences [34.5, 62.5].

All values 40–55 lie inside the fences. 90 lies outside ($90 > 62.5$).

Final Answer: 90 is an outlier.

Question 5: Determine the direction of skewness.

If Mean > Median → right-skewed (positive skew); If Mean < Median → left-skewed (negative skew)

Step 1: Compare mean and median found above.

$$\text{Mean} = 51.3, \text{Median} = 47$$

Step 2: Since Mean > Median, apply the right-skew rule.

$$51.3 > 47$$

Final Answer: The data is right-skewed (positively skewed), caused by the high outlier value 90.

Case Study 10 — Comparing Two Datasets

Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

SOLUTION:

Given Data

Dataset A, $n = 5$: 10, 20, 30, 40, 50

Dataset B, $n = 5$: 28, 29, 30, 31, 32

Question 1: Calculate the mean of both datasets.

$$\text{Mean} = \Sigma xi / n$$

Step 1 (Dataset A): Add all values.

$$10 + 20 + 30 + 40 + 50 = 150$$

Step 2 (Dataset A): Divide by $n = 5$.

$$\text{Mean}(A) = 150 / 5 = 30$$

Final Answer: Mean of Dataset A = 30

Step 1 (Dataset B): Add all values.

$$28 + 29 + 30 + 31 + 32 = 150$$

Step 2 (Dataset B): Divide by $n = 5$.

$$\text{Mean}(B) = 150 / 5 = 30$$

Final Answer: Mean of Dataset B = 30

Question 2: Calculate the variance of both.

$$\sigma^2 = \Sigma (xi - \mu)^2 / n$$

Step 1 (Dataset A): Find deviation and squared deviation of each value from $\mu = 30$.

xi	xi - μ	(xi - μ) ²
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10	-20	400
20	-10	100
30	0	0
40	10	100
50	20	400

Step 2 (Dataset A): Sum the squared deviations.

$$\Sigma(x_i - \mu)^2 = 400 + 100 + 0 + 100 + 400 = 1000$$

Step 3 (Dataset A): Divide by $n = 5$.

$$\sigma^2(A) = 1000 / 5 = 200$$

Final Answer: Variance of Dataset A = 200

Step 1 (Dataset B): Find deviation and squared deviation of each value from $\mu = 30$.

x_i	$x_i - \mu$	$(x_i - \mu)^2$
28	-2	4
29	-1	1
30	0	0
31	1	1
32	2	4

Step 2 (Dataset B): Sum the squared deviations.

$$\Sigma(x_i - \mu)^2 = 4 + 1 + 0 + 1 + 4 = 10$$

Step 3 (Dataset B): Divide by $n = 5$.

$$\sigma^2(B) = 10 / 5 = 2$$

Final Answer: Variance of Dataset B = 2

Question 3: Calculate the standard deviation of both.

$$\sigma = \sqrt{\sigma^2}$$

Step 1 (Dataset A): Take the square root of the variance.

$$\sigma(A) = \sqrt{200} = 14.14$$

Final Answer: SD of Dataset A = 14.14

Step 1 (Dataset B): Take the square root of the variance.

$$\sigma(B) = \sqrt{2} = 1.41$$

Final Answer: SD of Dataset B = 1.41

Question 4: Which dataset has greater spread?

Step 1: Compare the two standard deviations.

$$\sigma(A) = 14.14 \quad \text{vs} \quad \sigma(B) = 1.41$$

Step 2: The larger standard deviation indicates greater spread.

$$14.14 > 1.41$$

Final Answer: Dataset A has far greater spread.

Question 5: If both datasets represent model errors, which model appears more consistent?

Step 1: Consistency corresponds to lower standard deviation (less variation in errors).

$$\sigma(B) = 1.41 \text{ is much smaller than } \sigma(A) = 14.14$$

Final Answer: The model represented by Dataset B is more consistent.

Question 6: Explain why mean alone is not enough to compare datasets.

Step 1: Compare the means of the two datasets.

$$\text{Mean}(A) = 30 = \text{Mean}(B)$$

Step 2: Compare the standard deviations of the two datasets.

$$\sigma(A) = 14.14 \neq \sigma(B) = 1.41$$

Step 3: Conclude what the mean fails to capture.

Although both datasets have identical means, their variability is completely different, so mean alone cannot distinguish a highly-spread dataset from a tightly-clustered one.

Final Answer: Mean alone is not enough; variance/standard deviation must also be examined to fully compare two datasets.